



STORES Department
SIMHADRI

Ref:SMTTP/STORES/2022-23/12345678/986877

Date:31/03/2023(12:38:18)

Subject:LMI For 2023-24

Subject: Revision of LMI on Preservation of Materials in Stores”

LMI for “Preservation of Materials in Stores”

vide reference: LMI/01/C&M/MATERIALS/PRESERVATION, ISSUE NO. 01/REVISION NO.1/MAR 2023 has been reviewed and revised.

Following changes have been done in this Rev. No.1.

1. Entire document checked and corrected. Existing information in COS-ISO-00-OGN/OPS/SYST/017 Stores Management System Manual and Guidelines on Material Preservation is deleted to remove
2. Guide lines for nomination and operation of Preservation committee added.

- Types of Materials and scope of coverage during Preservation Committee visits added.

1. Format for Observations by committee and compliance procedure added.
2. Electronic Approval of LMI in PRADIP file management system has been incorporated.
3. Procedure for distribution of LMI has been revised as following: “Soft copy of the LMI shall be uploaded in the LAN by the Issuing Authority through a

Link edit protected through password. The uploaded soft copy of the document is to be considered as Controlled Copy. Copy of the document can be distributed through E mail”

Submitted by:

मधु बाबू आर्वा - उप प्रबंधक

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On:31/03/2023(12:38:18) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Reviewed , may be approved.

Submitted by:

देबजीत कुमार कर - अपर महाप्रबंधक

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On:31/03/2023(16:47:30) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Submitted by:

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On:31/03/2023(17:09:29) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

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All Maintenance Deptt. HODs are requested to kindly study the proposed revision of the LMI (see Annexure) and approve the same. Any changes if required, shall be recorded and case file sent to Central stores for the necessary revision in the proposed LMI.

Please note that all preservation work shall be done by Central stores as per recommendations of user departments, who shall carryout periodic vist of stores for checking the preservation of spares stored therein.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:01/04/2023(10:47:16) **Ref:**SMTPP/STORES/2022-23/12345678/986877

Pl review the document in consultation with all Groups and return back with comments if any for changes/correction

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

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On:01/04/2023(11:45:06) **Ref:**SMTTP/STORES/2022-23/12345678/986877

Following points may be added w.r.t. C&I Spares:

1. Storage of Accumulator Bladders:

- It should be kept in AC room, in hanging position.
- Folding is to be completely avoided to prevent deformation.
- Pressure inside bladders should be kept > 0.1 bar

2. Rubber Seal kits/O-Rings to be kept in AC room (in original packing).

Put forward for kind recommendation

Submitted by:

प्रदीप कुमार मौर्य - वरिष्ठ प्रबंधक

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On:04/04/2023(10:09:00) **Ref:**SMTTP/STORES/2022-23/12345678/986877

As discussed contents of latest OGN-OPS-SYST-017_Rev01_Long Term Storage of Power Plant and Equipment attached with this note to be followed.

Submitted by:

जी एच राव - अपर महाप्रबंधक

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On:04/04/2023(10:37:57) **Ref:**SMTPP/STORES/2022-23/12345678/986877

As per latest [OGN on long Term Storage of Power Plant Equipments](#), w.r.t C&I spares:

- Rubber items like seals, diaphragms, O-rings etc. should be stored in AC store. (clause 4.15 & Clause 11 of OGN).

Submitted for kind recommendation.

Submitted by:

प्रदीप कुमार मौर्य - वरिष्ठ प्रबंधक

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Reviewed and put up for approval please.

Submitted by:

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On:04/04/2023(23:02:43) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

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On:05/04/2023(10:26:41) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Review and comment if any.

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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On:05/04/2023(16:23:41) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Please include in Annexure-1 Sl.No. 13 "Control and Instrumentation" the following electronic spares of EMD-CHP, to be kept in AC room:

1. VFD/DC drive spares of M/s Siemens
2. Proximity sensors
3. PLC card spars of M/s Schneider & M/s Siemens
4. Pull chord controllers of M/s Parametrics
5. ELECTRONIC SPEED MONITORING UNITs & Speed sensor probes of M/s Jayashree
6. Spares of UPS, Battery chargers.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

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On:06/04/2023(09:31:25) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

May Review

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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On:08/04/2023(15:45:27) **Ref:**SMTTP/STORES/2022-23/12345678/986877

reviewed For HT motors and Transformers , found in order , forwarded for further approval

Submitted by:

सतीश कुमार सिंह - उप महाप्रबंधक

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On:08/04/2023(16:03:29) **Ref:**SMTTP/STORES/2022-23/12345678/986877

In LMI annex 1, item 13 and 22, EM may be added.

Preservation of rubber items (item 22).

Rubber items should be stored in AC room by applying a coat of talcum powder on the cords and gaskets.

Preservation of electronic items (item 13).

The philosophy of preserving the electronic cards is same as being done for C&I items.

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

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On:12/04/2023(09:47:41) **Ref:**SMTPP/STORES/2022-23/12345678/986877

Unauthorized Use Prohibited

Electronic cards & Rubber items like seals, diaphragms, O-rings etc. should be stored in AC store.

E-pac actuators consists of Electronic cards should be stored properly.

During storage and transportation in case of motor operated valves cover the actuator with a polythene bag.

Gear Boxes and Motors (Closed Storage P - Type)

- Shall be kept resting on its base.
- No external loading is permitted. Exposed shaft ends are to be covered with rust preventive coat and covered with polythene bags.
- Rotors are to be rotated periodically once in six months to avoid sagging and bearing jamming due to drying of grease.
- Terminal boxes of the motor shall be kept in sealed condition.
- Gear box should be kept filled with suitable oil up to working level to avoid rusting.
- Silica gel / moisture absorbent (Silica gel BS 2540) is to be kept along the motor and terminal box. Periodically silica gel to be checked / replaced or regenerated.

Lubricating and Cooling Oils (Semi Closed Storage - Q type)

- Drum / containers to be kept in horizontal condition with smaller cap on top.
- To be kept in sealed condition on hard surface / platforms.
- Keep away from the fire and inflammable type materials.

Reviewed for LT swgrs, Motors and actuators.

Forwarded for further

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

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On:13/04/2023(10:43:06) **Ref:**SMTPP/STORES/2022-23/12345678/986877

Forwarded for implementing the noting of DGM,EMD in LMI as per pre pages

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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On:13/04/2023(20:08:45) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Submitted by:

मधु बाबू आर्वा - उप प्रबंधक

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On:15/04/2023(09:24:21) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

BMD comments:

Following is proposed for inclusion in Annexure 1

Point no. 6 - Fan Assembly Gear Boxes & other critical items to be kept in Sheds.

Point no. 8 - All HP Boiler tubes to be kept in Sheds

Point no. 9- a.- Rust preventive coating to be done on threaded portions also

b. Critical spares like Bowl hub cover, Vertical shaft and Bowl extension cover to be kept in sheds.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:17/04/2023(10:07:57) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Review

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

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On:17/04/2023(16:50:32) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Reviewed LMI for Turbine rotors and Gear Boxes and found to be in order.

Forwarded for further processing.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

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On:19/04/2023(17:55:36) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Reviewed & forwarded for approval, please.

Submitted by:

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SITE)

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On:21/04/2023(17:59:54) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Review & comment if any.

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

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On:21/04/2023(19:07:34) **Ref:**SMTTP/STORES/2022-23/12345678/986877

Reviewed, forwarded for approval.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

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On:26/04/2023(11:08:16) **Ref:**SMTTP/STORES/2022-23/12345678/986877

Forwarded for approval please

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

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On:26/04/2023(11:17:40) **Ref:**SMTTP/STORES/2022-23/12345678/986877

Please confirm whether all the points of end users have been incorporated in LMI or not.

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

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On:01/05/2023(17:24:45) **Ref:**SMTTP/STORES/2022-23/12345678/986877

sir , All the user dept incorpoted all the points covered related to the user Dept .

Submitted by:

मधु बाबू आर्वा - उप प्रबंधक

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On:02/05/2023(09:37:00) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

May please be approved

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

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On:02/05/2023(10:27:42) **Ref:**SMTTPP/STORES/2022-23/12345678/986877

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

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On:03/05/2023(07:21:54) **Ref:**SMTPP/STORES/2022-23/12345678/986877



C&M-STORES Department
SIMHADRI

**LMI/01/C&M/MATERIALS/PRESERVATION
ISSUE NO. 01/REVISION NO.2/MAR 2023**

LOCATION MANAGEMENT INSTRUCTION

Preservation of Materials in Stores

**LMI/01/C&M/MATERIALS/PRESERVATION
ISSUE NO. 01/REVISION NO.2/MAR 2023**

Online Approval for Implementation by: Sanjay Kumar Sinha,
CGM (SIMHADRI)

E Office Ref No.: SMTTP/STORES/2020-21/AB.LMI20/157989

NTPC Ltd.
SIMHADRI SUPER THERMAL POWER STATION

**LMI/01/C&M/MATERIALS/PRESERVATION
ISSUE NO. 01/REVISION NO.2/MAR 2023**

LMI APPROVAL DOCUMENT

1	TITLE	Preservation of Materials in Stores	
2	Document Code	COS-ISO-00-OGN/OPS/SYST/017	
	Revision /Issue No. & Date	29.03.2023	
	Name	Designation	Signature
Prepared by	Sh. Arava Madhubabu	Dy-Manager (Stores)	
Reviewed by	Sh. Debjit Kumar Kar	AGM(Stores)	
Reviewed by	Sh. Mathew Eipe kovoov	GM(Maintenace)	
Reviewed by	Sh. Dhrjati Prasad Patra	G.M(O&M)	
Approved by	Sh. Sanjay Kumar Sinha	CGM(Simhadri)	

NTPC Ltd.
SIMHADRI SUPER THERMAL POWER STATION

**LMI/01/C&M/MATERIALS/PRESERVATION
ISSUE NO. 01/REVISION NO.2/MAR 2023**

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Preservation of Materials in Stores

1.0 INTRODUCTION:

This LMI deals with the procedure to be adopted at NTPC Simhadri, regarding the storage and preservation of spares and consumables. This document is prepared in line with Corporate T.C. Document COS-ISO-00-OGN/OPS/SYST/017 (with revisions), Stores Management System Manual and Guidelines on Material Preservation, Material Handling, Safe Operations and Construction of Storage Spaces and Open Yards prepared and issued by corporate materials group.

2.0 SCOPE :

The Scope covers Storage type, preservatives to be applied, the methodology of inspection, and the responsibilities of different departments in the implementation of this LMI.

3.0 SUPERSEDED DOCUMENT:

COS-ISO-00-OGN/OPS/SYST/017

Clause 1.3: Review requirements changed to 2 years.

II. Clause 4.10 (previous 3.10): Preservation recommendations of Turbine casing, Turbine blades, Journal Bearing and All machined surfaces of turbine components like fasteners, coupling bolts was included additionally.

III. Sub Clause 4.15(Previous 3.15): Additional point “Ensure monthly visit by C&I Executives to ensure the proper preservation of these items” has been added. Long Term Storage of Power Plant and Equipment: General Considerations and Preservation COS-ISO-00-OGN/OPS/SYST/017 Rev. No.: 1 Dec. 2022 Page 3 of 37

IV. Clause 4.20 (previous 3.20) WT plant/Chemical Storage: for cation resin, to maintain moisture, add DM water only (previous normal water). Once in a month change water (earlier once in a Year).

V. In clause no 4.6, Preservation type for PA, FD and ID fan rotors added.

VI. In clause 4.8, Preservation type for Boiler tubes modified.

VII. Clause 4.20.2 (previous 3.20.2): Ammonia and hydrazine storage rooms must have exhaust fans for ventilation. Safety shower /Eye washer should be available near to chemicals storage area. These new clauses have been added.

VIII. New Clause 4.20.6 CW Treatment Chemicals (Closed Storage P-Type) was added.

IX. Clause 5.5.1 (previous 4.5.1): PT- Type: Closed Shed with Dry and Dust free atmosphere Type of Construction: Feature of AC room with respect to Type of Air conditioner, power supply provision, remote indication of temperature and humidity data has been added.

X. Type of Preservatives at Clause 6.2.1 (previous 5.2.1), preservatives TRP HE 1606, TRP HE 1706, TRP HE 1610, TRP HE 1710, TRP HE 1612, TRP HE 1712, Mobilux Grease have been added.

XI. Preservation type for Ash Handling spares (Bag filters / cage / vacuum pumps/ clinker grinders etc.) added at clause 4.21.

XII. Electronic cards & Rubber items like seals, diaphragms, O-rings etc. should be stored in AC store at clause 4.15

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XIII. PRESERVATION OF EHV CT/CVT/OIP BUSHING and Generator winding spare bars at clause 4.14

4.0 LIST OF REFERENCE DOCUMENTS:

(Available in Intranet, Corporate Website)

- a) COS-ISO-00-OGN/OPS/SYST/017 (with revisions)
- b)) Stores Management System Manual
- c) Guidelines on Material Preservation, Material Handling, Safe Operations and Construction of Storage Spaces and Open Yards

5.0 REVISION HISTORY:

- i. Entire document checked and corrected. Existing information in COS-ISO-00-OGN/OPS/SYST/017, Stores Management System Manual and Guidelines on Material Preservation is deleted to remove redundancy.
- ii. Guide lines for nomination and operation of Preservation committee added.
- iii. Types of Materials and scope of coverage during Preservation Committee visits added.
- iv. Format for Observations by committee and compliance procedure added.

6.0 Methodology of Inspection:

A Preservation Committee nominated by respective HODs and approved by HOS, generally consisting of following members shall ensure preservation of the materials, under the general guidance and directions of Head (O&M) and In-Charge- (Stores/C&M).

- a) In-charge-Custody (Stores)
- b) MTP representative (Convener of Stores Visits)
- c) FQA. Representative (whenever required)
- d) Representatives from MM, EM, TM, OS, AHP, CHP & MGR.

6.1 General Scope of Inspection is given in Annexure –I. Scheduled Visits and Inspections shall be carried by the committee. The observations made during the inspection shall be entered in the Register maintained

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in the stores. Corrective & Preventive actions shall be carried out in line with these observations.

6.2 This register shall be reviewed by In-Charge- (Stores/C&M) on monthly basis and Head(O&M) every quarter.

6.3 In-charge-Custody (Stores) shall raise Indents and Proposals for Materials and Works required for Preservation Activities of general and periodical nature. Specific Preservation activities related to Capital/Pooled Spares, Critical Spares & equipment of TMD, Electrical, C&I, Special Tools etc. shall be done through respective departments. This is required to ensure that no damage occurs through improper preservation method.

ANNEXURE -1 : General Scope of Inspection during Preservation Committee Visit:

S.N	Item Description	Store Type	Laydown/Storage Procedure	Insp. Interval	Inspecting By
1.	STEEL BEAMS/ ANGLES/ CHANNELS/ PLATES	'R'	a. Yard to be leveled even ground, free from grass/ weed and preferably brick paved b. Multiple supports at internals to avoid sagging. c. Flanges of sections to rest on supports (wherever applicable) d. Proper floor drainage shall be arranged. e. To store section / item wise, with clear naming.	03 Months	MM/CHP
2.	PIPES/TUBES/ HEADERS	R	a. Yard to be leveled even ground, free from grass/ weed and preferably brick paved. b. Multiple supports at internals to avoid sagging. c. Flanges of sections to rest on supports (wherever applicable) d. Proper floor drainage shall be arranged. e. To store section/item wise, with clear naming.	03 Months	MM/CHP

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3.	VALVES & ACTUATORS	P	a. Rust preventive coating on body. b. Placement upright. c. Greasing of unpainted surface with grease or preservative oil. d. Valves to be kept with disk/wedge closed (maintenance shall assist where ever required). e. Plugging of valve flanges/ends with PVC caps / (Maintenance/User Department shall assist where ever required). f. To ensure no oil leakage through gaskets/seals of valve actuators.	03 Months	MM/EMD/ CHP
4.	Wall De-slaggers	P	a) Water proof covering of blowers. b) Wall de-slaggers and rotor heads and other assemblies are supplied with temporary wooden platforms to prevent parts getting damaged during handling. These should not be removed unless used. c) All electrical items, nozzles and valve assemblies are openings are to be covered with water and dust proof coverings.	03 Months	EMD/BMD
5.	PUMPS/ SHAFTS/ IMPELLER ASSEMBLY/ GEAR BOX	P	a) To store in original frame/packing. b)Plugging of pumps ends. c)RP- oil coating/evaporative coating on exposed ends of shaft & impellers. d) 06 monthly rotation of shafts to avoid jamming. (Maintenance/User Department shall assist where ever required). e) impellers are to be coated with rust preventive coat. f)Shafts shall be coated with rust preventive coat on bearing portion & VCI paper. g)rotating elements shall be kept on stand to facilitate rotation. h) Rust preventive coat on body, hand wheel , flanges. Etc. i) Threaded portion to be coated with rust preventive coat.	03 Months	MM/CHP

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6.	FANS	RT	a. To be kept on block supports. b. Rust preventive coat on journals/bore surfaces. Suitable support for shafts/impellers (where ever feasible) to avoid jamming/sagging. Maintenance/User Department shall assist where ever required). c. Impellers to be kept with in vertical position. d. Rotation of shaft once in 6 months. e.Date of Rotation shall be indicated on the item.	03 Months	MM/CHP
7.	IMPELLERS, PEDESTALS	RT	a) Above ground on block supports- b) When kept outdoors these shall be covered with tarpaulin and vented for air circulation. A gap of minimum 300 mm shall be made available between the components and its covering.- c)Rust preventive coating must be there on the journals / other bare surfaces- d) Shaft and impeller assembly should be supported at suitable span to avoid sagging. e) Periodic rotation for heavy shafts/ rotors must be done. Impellers are to be kept with axis in vertical position. f) Impeller and shaft should be preferably kept indoors (P-type storage) along with bearing assembly, on the frames received from the shop. h)Check for RP coating if it is not there then apply the same	03 Months	MM/CHP
8.	BOILER TUBES	P	a Ends of tubes are to be capped. b. Primer coat on outside surface to be intact. c. Preserve color coding of tubes. d. Availability of moisture inhibitor pallets (silicagel BS 2540) along with coils.	03 Months	BMD

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9.	COAL MILL PARTS	RT	a) Suitable rust preventive coating (Class-A) must be there on the machine seating / matching surfaces. b)Laid on leveled ground on blocks. c) Rolls are to be kept in vertical axis when stacked (max. 3 nos.) d) Ring shall be kept horizontally and stacked (max. 3 nos.) e) Ring shall be kept horizontally and stacking shall not be more than 2 nos. on wooden sleepers with its gay ropes intact. f) No other external loading.	03 Months	BMD
10.	ROTORS OF TURBINE	P	a) .To keep in original packing. b). Journals to be coated with rust preventive material & covered with polythene bags/sheets. c). Inspection replacement of silica gel type moisture absorbents for motor rotors/stators. d). Rotation of rotors once in 6 months and lubrication of support bearings as required.	03 Months	TMD
11.	BEARINGS	P	a. Storage in original pickings / polythene bags cartons. b.To be kept in horizontal position. c. Inspection for rusting/corrosion and integrity of rust preventive coatings. d. In case of friction (journal) bearings parting planes, white metals and saddle seating faces to be coated with rust preventive oils / grease. e. No stacking of bearings in case of friction bearings.	03 Months	MM/CHP/E MD/MTP

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12.	GEAR BOXES	P	a. Exposed shaft to be coated with rust preventive coat and to be covered with polythene bags. b. Rotor to be rotated once in 6 months to prevent sagging and jamming due to drying of grease. c. Terminal box gland hole of motor should be plugged with PVC caps. d. Suitable oil level to be ensured in gearboxes(maintenance to mark normal oil level, indicate the oil type and also assist in oil top up). e. Silica gel to be kept along the motor and terminal Box. Silicagel to be periodically to be checked / replaced or regenerated	03 Months	CHP/BMD/TMD
13.	Control and Instrumentation	PT	a) Cards, electronics and microprocessor-based components shall be stored in AC rooms in their original packing. b) Silica gel (BS 2540) / moisture absorbent is to be kept around and replenishment is to be done periodically. c) Electronic Cards, complete peripherals and small items may be kept in anti-static charge polythene pouches. d) Critical relays, recorders and vulnerable electric spares parts may be kept in anti-static charge polythene pouches	03 Months	C&I

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14.	Welding Electrodes	P	a) To be stacked on steel racks / cupboards b) Stands made of moisture absorbent type material like wood, to be avoided. c) Carton / polythene bags shall not be opened / removed other than those required for random inspection during receipt. d) Cartons stacking is to be of minimum height so as to avoid the damage to bottom layer boxes and electrodes fluxes. e) No external loading on electrodes is permitted. f) Welding electrodes expiry dates are to be seen and informed to users well in advance after receipt of the same in batches. If the expiry date is not mentioned, then to be used within two years of manufacturing. g) Electrodes should be kept in hot air ventilated room free from moisture.	03 Months	MM
15.	Refractory Insulation Material	P	a. Mineral wool mattresses are to be kept above ground and stacking height shall not be more than 2 Mtrs. b. Pourable insulation bags to be stored in a dry place c. Refractory bricks are to be kept in semi- covered/open yard. When bricks are stored one over the other, the height should be restricted to avoid damage to the bricks in lower level. refractory Bags are t be kept inside the Godowns. While stacking bags, the height should be restricted to avoid damage to the Bags forming the lower level. power bags and special bricks should not be exposed to rain/water or damp condition and kept inside the Godown.	03 Months	AHP/BMD

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16.	Cement	P	a) Above ground in dry closed room. b) There should not be any ventilation in room. c) Shall be kept 6" away from the wall. d) While stacking the bags, the height should be restricted to avoid damage to the bags forming the lower level.	03 Months	Civil
17.	Resins	P	a) Resins to be stored in plastic drums. b) Resins drums to be stored under shade. c) Always maintain moisture in resin. d) For cation resin, add normal e) For anion resin, add DM water/.clean water. f) Once in a month, rotate the drums up and down g) Once in a year, change the water.	03 Months	Chemistry
18.	Activated carbons	P	a) To be preserved in original packing or in plastic drums. b) The packing should be completely covered to avoid dust. c) Storage area should be away from fire and inflammables.	03 Months	Chemistry
19.	Rubber Lined Pipes	P	Rubber lined pipes, vessels, polypropylene strainers, laterals and vessels are to be kept covered.	03 Months	AHP/MM-OS
20.	Industrial Gases	P	a) To rest on concrete pad or on combustible racks. b) Colour coding as per standards. c) Cylinder Hydro test validity date to be checked and ensured. d) Statutory requirements to be met. e) To store separately oxygen, acetylene & chlorine and to avoid mix up. f) Cylinder position : Acetylene – Vertical. g) Screw on cap on valves.	03 Months	MM

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21.	Bulk Chemicals	P	a) These bulk chemicals should be stored in separate rooms to avoid contact with one another b) Drums, jerry canes and bags should be placed up right on sleepers c) To keep in dry cool place away from direct sunlight	03 Months	Chemistry
22.	RUBBER SEALING, CORDS, MOULDED RUBBERPACKING AND JOINTING MATERIALS.	P	a. To be stored in dry, air tight containers, away from direct lighting. b. Contact with liquid / semi liquid solvents, oil, grease, etc., to be avoided. c. FIFO to be followed strictly. d. Aged, unserviceable material to be segregated and disposed as per NTPC systems.	03 Months	CHP/MM
23.	PAINTS / VARNISHES/SOLVENTS/THINNERS	P	a) To store in Cool, dry place b) To store in leak free, rust proof air tight containers. c) FIFO to be ensured.	03 Months	MM/EMD/CHP
24.	HT / LT Motors	P	1)Preservatives to be applied on the projected portion of shaft. 2) 6 monthly rotation of shafts (by maintenance) 3) Yearly meggering of the motor windings(By maintenance). 4) For HT motors, space heater connection to be given and switched 'ON'. (By maintenance).	03 Months	EMD
25.	Cable drum	RT	a) To store on concrete paved floor. b) Axis of the drum should be horizontal. c)End should be capped with PVC cap. d)Sufficient space between drums.	03 Months	EMD/C&I
26.	Transformer bushings	R	a)To keep vertical (To fabricate special stands if required) b) Tan delta once a year	03 Months	EMD

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27.	LUBRICANTS	Q	a)Storage within the temperature limits extreme cold (below -15 degree C) and extreme heat (above 45 degree C). b)Storage in easily accessible, well ventilated and well lighted, spacious, away from heat sources and well protected from rains. Out Door Storage Method: a) Drums to be placed on wooden planks in cool indoor area. b) Layers limited to two/three numbers. c) Kept in horizontal position with the bungs at 3 O'clock and 9'O clock position. d) Direct sun rays should not fall on drums. Indoor Storage Method: a)Clean from industrial contamination. b)Sufficient space between drums for easy movement. c)Grease drums should be kept in upright Position to avoid spillage. Shelf Life Of Oils and Greases: <table><tr><th>Product</th><th>Shelf Life in Years</th></tr><tr><td>Base Oil –</td><td>Above 5</td></tr><tr><td>General Lub. Oils-</td><td>5</td></tr><tr><td>Greases-</td><td>2-5</td></tr><tr><td>Soluble Cutting Oils-</td><td>1-5</td></tr><tr><td>Steel Mill Rolling Oils-</td><td>2-5</td></tr><tr><td>Rust Preventive-</td><td>2</td></tr><tr><td>Open Gear Oils-</td><td>2-5</td></tr><tr><td>Compounded Oils-</td><td>2-3</td></tr><tr><td>High Performance Ind. Oils-</td><td>3-5</td></tr><tr><td>High Performance engine Oils-</td><td>3-5</td></tr></table>	Product	Shelf Life in Years	Base Oil –	Above 5	General Lub. Oils-	5	Greases-	2-5	Soluble Cutting Oils-	1-5	Steel Mill Rolling Oils-	2-5	Rust Preventive-	2	Open Gear Oils-	2-5	Compounded Oils-	2-3	High Performance Ind. Oils-	3-5	High Performance engine Oils-	3-5	03 Months	BMD/TMD/ CHP
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NOTE: For other items /spares not covered, the guide lines given in COS-ISO-00-OGN/OPS/SYST/017 (with revisions if any) and the manufacturer's guide lines for storage, catalogues shall be referred as required.																											
TYPE OF STORAGE:- 'P' Type - Fully closed 																											

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ANNEXURE 2 : Format of Inspection Observation Register

Sl. No	Date & Dept. Name	Inspected by Name/Design	Material Code of Items/Items inspected	Location of Item	Observations if any.	Signature & Date	Action taken by Stores	Remarks by I/C(Stores)

Review Period : In-Charge- (Stores/C&M) on monthly basis and Head(O&M) every quarter.

7.0 DISTRIBUTION LIST

1. Simhadri INTRANET

8.0 REVIEW FREQUENCY & RECORD

REVIEW INTERVAL: This LMI will be reviewed by In-Charge (STORES/C&M) once in two years or as and when required.

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